Let's review SynthSeg+ methods, plz refer my uploaded papers (https://www.pnas.org/doi/full/10.1073/pnas.2216399120#bibli… and link (https://github.com/BBillot/SynthSeg). I wanna know what is the preprocessing does the pipeline to do before segmentation?

SynthSeg- Segmentation of brain M…
5.38 MB

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Searching: SynthSeg+ preprocessing PNAS 2216399120

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Searching: SynthSeg+ preprocessing pipeline

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Below is a consolidated, practical description of the preprocessing performed (or intentionally avoided) in the SynthSeg and SynthSeg+ pipelines before running the core segmentation, synthesizing information from the original SynthSeg paper you supplied plus publicly available SynthSeg / SynthSeg+ resources. I separate: (A) what the method itself internally expects or does at inference; (B) what the FreeSurfer / command-line wrappers may add; (C) what SynthSeg+ extends; and (D) optional vs discouraged external preprocessing.

A. Core (minimal) preprocessing actually required by SynthSeg

SynthSeg was explicitly designed to work with virtually no mandatory external preprocessing. The network's robustness arises from domain randomization during training (random orientations, scales, intensity profiles, contrast, resolution, bias fields, presence/absence of skull, lesions, etc.). Consequently, at test time only two lightweight steps are strictly performed:

1. Resampling to an internal target isotropic resolution (typically 1 mm):
    - The input scan (MRI or CT) is interpolated (trilinear) to 1 mm isotropic voxel size so that the CNN always operates at a fixed grid.
2. Intensity normalization (simple min–max or percentile scaling):
    - Intensities are linearly rescaled to [0, 1]; in the published implementation this uses (often) the 1st and 99th percentiles to reduce outlier impact, then min–max normalization.

Everything else (bias correction, skull stripping, denoising, orientation standardization) is optional because the network learned to cope with their presence or absence. The model was trained with: random rotations (so orientation does not need correction), random inclusion/exclusion of extra-cerebral tissue (so skull stripping not required), random multiplicative smooth bias fields (so N4 / bias correction not required), gamma/intensity warps (so histogram matching not required), and synthetic low-resolution simulation (so slice thickness / spacing variability is tolerated). [3] [4]

B. What the FreeSurfer / released tools may still do around SynthSeg

When you run SynthSeg via FreeSurfer (e.g., mri_synthseg):

- Conformation / geometric standardization: Depending on input, a "conform" step (common in FreeSurfer utilities) may reorient to RAS, set voxel size, and (if needed) pad or crop; SynthSeg itself only needs consistent voxel size, not a specific orientation, but the wrapper may standardize for consistency with other FS tools.
- Optional skull stripping flags: Even if you pass a skull-stripped volume, the network behaves well because training randomly dropped extra-cerebral labels 50% of the time.
- For CT: HU clipping and rescaling (e.g., clipping to a plausible window such as [-1000, ~1000] HU and linear scaling to [0,1]) may be applied in wrapper scripts; the network then proceeds identically.

- Test-time augmentation: A left–right flip pass may be performed and averaged (ensembling) to slightly improve stability (mentioned in the original paper).

These wrapper steps are convenience measures, not strict requirements; the robustness stems from the training strategy. [3] [4]

C. Additional preprocessing / orchestration introduced or leveraged in SynthSeg+

SynthSeg+ (PNAS "Robust machine learning segmentation…"), the extended suite, layers additional components on top of the core SynthSeg segmentation to support large-scale heterogeneous clinical data analysis (e.g., volumetrics, cortical thickness, QC). Key practical preprocessing or quasi-preprocessing elements include:

1. Unified lightweight normalization for any clinical input:
   - Same minimal resample + intensity scaling paradigm retained. Heavy pipelines (e.g., full FreeSurfer recon-all) intentionally avoided for scalability. [1] [2]
2. Super-resolution–aware shape inference (implicit):
   - Instead of explicit multi-slice super-resolution preprocessing, SynthSeg+ relies on the base network's learned ability to "super-resolve" thin structures (cortex) from thick-slice clinical scans after simple resampling. No explicit external SR pre-step is required. [1] [4]
3. Direct surface / cortical thickness derivation from segmentations:
   - Thickness estimation is driven from the segmentation (label-based morphology) rather than intensity-gradient surface finding, reducing need for bias correction, intensity non-uniformity correction, or topology repair steps that classical surface pipelines demand. This shifts what would traditionally be "preprocessing" into robust post-segmentation modeling—thus simplifying inputs. [1] [2]

4. Intracranial volume (ICV) and volumetric QC:
    - ICV is derived directly from the segmentation labels (no extra masking step).
    - QC / outlier detection uses standardized volumetric metrics—no additional preprocessing beyond the base normalization. [1]
5. Lesion / pathology robustness:
    - Because lesion-like patterns were synthetically injected during original training (random lesion "pasting"), SynthSeg+ does not mandate explicit lesion filling or WMH preprocessing. [4]

Overall, SynthSeg+ deliberately preserves the "minimal-in / maximal-out" preprocessing philosophy: enforce consistent voxel size + simple intensity scaling, rely on learned invariances for everything else. [1] [2]

D. Optional external preprocessing: When (and when not) to use

Although not required, you might consider these in edge cases:

Optional (potential marginal gains in extreme scenarios):

- Denoising (e.g., Rician noise filtering) if input is extremely noisy (e.g., old low-field scans). Usually the model already tolerates typical clinical noise.
- Gross artifact removal (truncate extreme spikes, correct severe wrap-around) if scanner artifacts are severe.
- Motion correction across 3D stacks if you only have multi-slab or multi-echo compositing issues (rare for standard single structural).
- CT windowing: If atypical acquisition ranges, standardize to a soft-tissue window before scaling to [0,1].

Generally Discouraged / Unnecessary:

- Histogram matching to a “template”: the domain randomization already encompasses diverse intensity distributions.
- Bias field correction (N4) as a default: synthetic bias augmentation makes it redundant in most cases.
- Skull stripping prior to SynthSeg: can introduce failure points; only do it if you need a masked brain for another downstream tool.
- Rigid/affine registration to a template before running SynthSeg: not needed for segmentation accuracy (only for group analysis after).
- Aggressive resampling above native resolution (>1 mm) hoping for more detail: no real new information; 1 mm isotropic target is sufficient. [3] [4]

E. Summary “Minimal Pipeline Recipe”

Given a raw clinical structural MRI (any orientation, contrast, slice thickness, or a head CT):

1. (Load) Read image as-is.
2. (Resample) Trilinear resample to 1 mm isotropic (no need to crop skull).
3. (Intensity scale) Clip (optional mild percentile clipping) and linear scale to [0,1].
4. (Run SynthSeg / SynthSeg+) Obtain soft label maps.
5. (Optional augmentation) Average with a left–right flipped inference.
6. (Post) Compute volumes (including ICV), surfaces/thickness (SynthSeg+), QC metrics.

Everything else (bias correction, skull stripping, histogram matching, lesion filling, etc.) is optional and usually omitted.

[1] [2] [3] [4]

F. Practical Notes / Caveats

- Extremely anisotropic scans (e.g., 1×1×8 mm) are still merely resampled; model's learned priors handle through its synthetic low-resolution training exposure. [4]
- For CT, ensure correct physical spacing meta-data; mis-specified DICOM spacing can mis-scale volumes (preprocessing should verify header fidelity). [3]
- If integrating into a batch pipeline, track the original affine for later group registration; SynthSeg itself does not require alignment to a common atlas but downstream statistical analyses do. [1]
- Quality control: leverage the volumetric outlier detection in SynthSeg+ rather than adding external heuristic brightness checks. [1] [2]

G. Difference in Philosophy (SynthSeg vs. Traditional Pipelines)

Traditional structural pipelines invest heavily in preprocessing (skull stripping → bias correction → intensity normalization → topology correction) to enable subsequent model-based segmentation. SynthSeg inverts this: it invests in a broad synthetic training distribution so that almost all those steps become unnecessary at inference, collapsing preprocessing to two cheap, deterministic operations (resample + scale). SynthSeg+ preserves that minimalism while expanding outputs (thickness, QC, multi-contrast applicability). [1] [3] [4]

If you need:

- A schematic flowchart of just the steps relevant to plugging SynthSeg+ into an existing repository
- Comparison with a legacy FreeSurfer recon-all preprocessing chain Let me know and I can provide those.

Cited Sources

[1] PNAS SynthSeg+ article (Robust machine learning segmentation for

large-scale analysis…)

2 PubMed entry for the same SynthSeg+ study

3 FreeSurfer SynthSeg Wiki (tool behavior & usage notes)

4 Original SynthSeg (MedIA / preprint resources)

Let me know if you would like more granular implementation details (e.g., exact percentile values, internal patch sizes, or how surfaces are derived in SynthSeg+).

Try web version for PDF details/comparison